CHAPTER

ATA-38

WATER AND WASTE SYSTEM

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Contents

Water/Waste - General

Introduction

The water and waste system are comprised of the following sub-systems:

- Wash water
- Waste water

The lavatory wash water system supplies the water for the-passenger washroom facilities on board the aircraft.

The system also-stores the waste water after passenger use and allows it to be-removed from the aircraft through a service panel.

General Description

The lavatory water system allows passengers to wash their hands in-the lavatory with warm water. All components are installed below the-sink in the lavatory compartment.

Servicing is done through a service panel installed on the exterior fuselage. The dc electrical system-supplies power to the switches, valves and sensors which operate-the lavatory water system. The ac electrical system supplies the-power necessary to heat the system. The lavatory waste system is an electronically operated, recirculating-flush toilet installed in the lavatory compartment.

The lavatory-compartment is manufactured in one piece. Any fluids spilled on to-the floor are contained within the compartment and then drained-overboard through the floor drain. The toilet unit is filled with clean-water and is drained of waste water through an exterior service panel.

Wash Water

Introduction

The wash water system supplies warm water to the lavatory sink for use by the passengers and crew.

General Description

The lavatory water system allows passengers to wash their hands in the lavatory with warm water. All components are installed below the sink in the lavatory compartment. Servicing is done through a service panel installed on the exterior fuselage

The DC electrical system supplies power to the switches, valves and sensors which operate the lavatory water system. The AC electrical system supplies the power necessary to heat the system.

On aircraft with SB84−25−159 incorporated and without SB84−25−160, the warm water wash system is deactivated.

On aircraft with SB84−25−165 incorporated, all the components related to the warm water wash system are removed.

On aircraft with SB84−24−59 incorporated and without SB84−24−61, the drain mast is deactivated

Detailed Description

Refer Figure 38-1 & *Refer Figure 38-2*

The lavatory water system has a 27 L water storage tank and a 1.7 L heater tank. An electric motor driven gear pump supplies system pressure. The system has one faucet with a built in switch which activates the pump.

The temperature of the hot water is approximately 34 °C. All water lines and components are heated by electric heating elements.

Waste water is drained from the sink directly overboard through a drain mast installed in the underside of the aircraft. There is also a manual drain operated from the external service panel. This allows ground crews to drain the system for overnight storage in freezing conditions.

A ground service cart is used to fill the 27 L storage tank. The supply line from the cart is attached to the fill adapter on the external service panel. The T−handle on the panel is turned 90 degrees in a clockwise direction. The T−handle is connected to the valve control unit located just behind the service panel. The control unit changes the turning movement of the T−handle to the linear movement of a cable which opens the filler valve.

Once the valve is opened, water flows into the fill line for the tank. A flow restrictor in the fill line to the tank makes sure there is no increase in pressure in the tank during the filling process. The tank has a pressure relief valve to prevent overpressure. A check valve in the fill line to the heater prevents any reverse flow from the heater to the storage tank. As the tank fills, the water opens the check valve allowing water to fill the heater vessel. When the tank is approximately 90% full, water flows out the overflow line and drains through the drain mast. The overflow is visible to the servicing crew and indicates that the tank is full.

A switch and a control box are located near the faucet. After 8 minutes, push and hold the faucet switch. Make sure that warm water comes out of the faucet and the water pump operates while you push the faucet switch

Waste water from the sink is dumped overboard through the drain mast installed in the underside of the aircraft. An airstop valve in the sink drain line is opened by the force of the waste water.

When the water stops flowing, the airstop valve closes to prevent cabin pressure from venting through the drain during flight.

A filter installed upstream of the pump prevents foreign objects from entering the pump and contaminating the system.

The heater uses a thermostat controlled electric blanket to maintain the water temperature at approximately 34 °C. A thermal fuse limits the water temperature to a maximum of 60 °C, in the event of a control thermostat failure.

A solid state low level switch shuts off the heater blanket and the pump when the water level is too low. The low level switch also turns on the LOW LEVEL POWER CUT−OFF light located on the lavatory control panel below the sink.

Heater cuffs protect the fill and drain valves, the pressure relief valve, the filter, the pump, the check valve and the tank from freezing during flight. The cuffs are either molded directly to these components or are fastened by bolts. The drain mast and water lines have built in heaters.

All the heaters have thermostats which control the temperature in the individual components. The heater system is energized as soon as AC power is connected to the aircraft. This prepares the system for filling by preventing flash freezing of cold water coming into contact with cold components.

The pump and control relay are powered from the 28 V AC RIGHT SECONDARY BUS through a 5 A LAV SYSTEM circuit breaker. The heaters are powered from the 115 V AC VARIABLE FREQUENCY RIGHT BUS through a 10 A LAV HTRS circuit breaker.

On aircraft with SB84−25−159 incorporated and without SB84−25−160, the LAV SYSTEM (D3) circuit breaker located on the 28 V DC right circuit breaker panel and the LAV HTRS circuit breaker located on the 115 V AC variable frequency circuit breaker panel are removed.

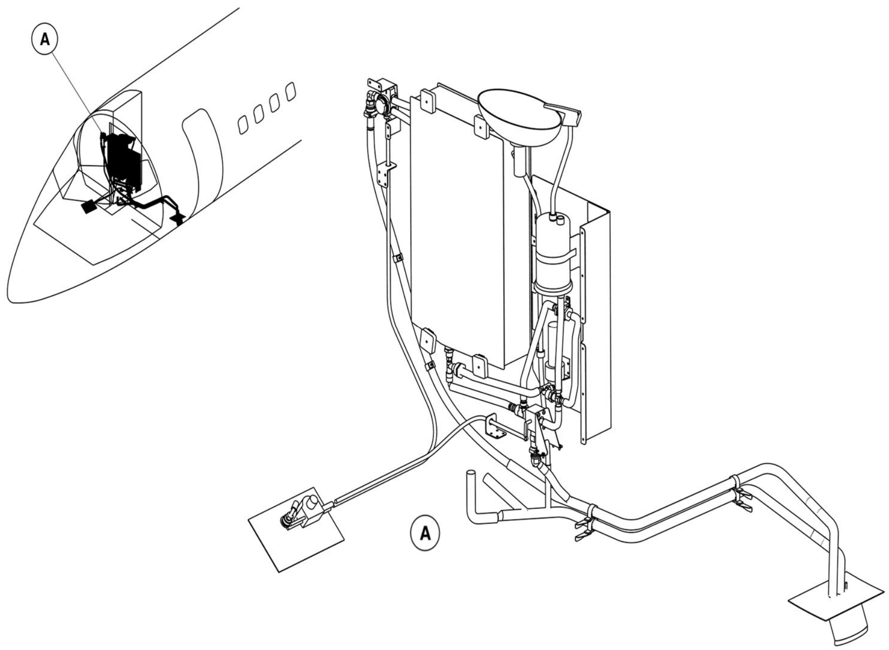


Figure 38-1 Lavatory Wash Water System – Location

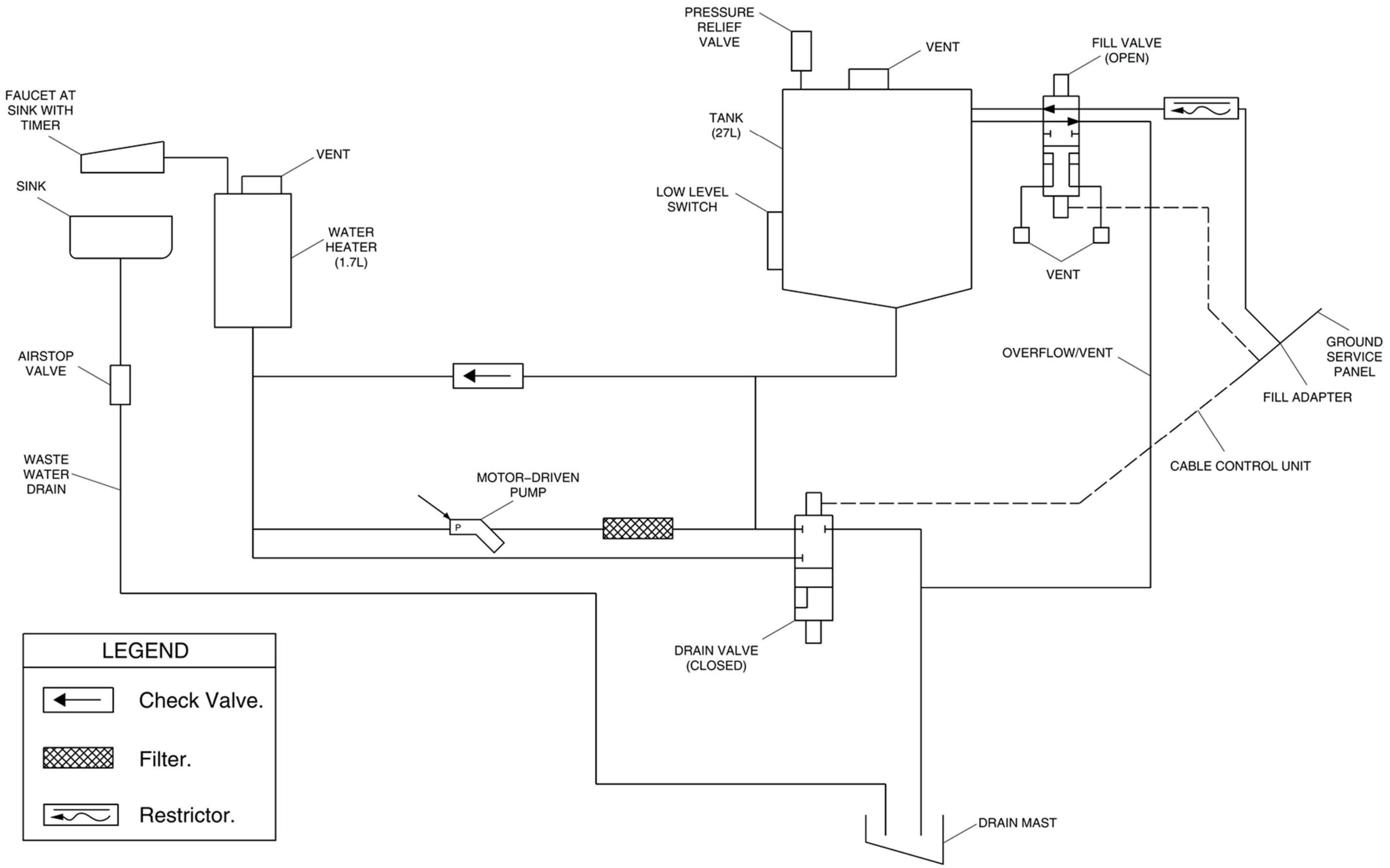


Figure 38-2 Lavatory Wash Water System – Schematic

Wash Water - Component Description

Refer Figure 38-3

Tank

The water tank provides storage for 27 litres of water, and provides the pump with positive head pressure.

The tank is installed above the floor, in a compartment under the lavatory sink. It is made from welded stainless steel. On the face of the tank, there is an access hole to allow for cleaning by hand. A cover and gasket is attached with 8 fasteners, sealing the hole.

Fill Valve

The heated fill valve is a 4 way, 2 position valve. The valve is mechanically operated from the rotary control box located at the ground service panel.

In the open position, the valve allows water to flow from the fill line to the tank. It also allows overflow water from the tank to flow through to the drain mast.

In the closed position, the valve isolates the tank from outside ambient pressure. Two internal vents in the valve break the vacuum in the fill and overflow lines allowing water trapped within the valve to drain away.

Check valve

A check valve is installed in line between the tank and the heater. The head of water in the tank forces the water through the check valve into the heater during the filling process. The check valve prevents the water in the heater from flowing back into the tank.

Heating Element

A heating element with a built in thermostat is molded directly on to the valve body. The heater uses 115 V AC variable frequency and the temperature is controlled between 10 and 30 °C.

Pump

The pump is a 28 V DC geared motor pump, with overtemperature protection. A magnetic clutch connects the motor to the gear drive. At a specified torque, the clutch disconnects and allows the motor to free spin.

The overtemperature protection is a self-resetting thermostat located in the motor housing. The thermostat switch turns off the electrical power supply if the motor overheats. The pump is protected from freezing by an external heating cuff installed on the pump housing.

The pump has an internal bypass valve which prevents the system from over pressurizing during operation. The bypass valve allows water to recirculate from the valve outlet back to the inlet. The bypass pressure is set at 38 psi (262 kPa), but is adjustable.

Drain Valve

The drain valve is a 2 position, 3 way ball valve, mechanically actuated from the rotary control box located at the ground service panel. The drain valve allows the entire system to be drained to prevent freezing in cold temperatures.

A heating element is molded to the top of the valve. A built in thermostat controls the temperature of the heating element between 10 and 30 °C. The heater element uses 115 V AC variable frequency power.

Filter

The filter is sintered stainless steel woven wire mesh inside a stainless steel case. It has a nominal rating of 40 microns, 75 microns absolute. It removes contamination from the water upstream from the pump. This prevents damage to the pump or clogging of other parts of the system. The filter is reusable and can be cleaned by backflushing or ultrasonic cleaning. The filter is heated by an external heater cuff.

Heater

The heater is a 1.7 L stainless steel tank. A 450 W heating blanket molded to the heater supplies heating for the water in the tank. A built in thermostat maintains the water temperature at about 34 °C.

If the thermostat fails, a thermal fuse will melt and turn off the heater when the water temperature reaches 60 °C. This protects the passengers against scalding water.

When the faucet lever is pushed down, a cartridge in the faucet opens and a microswitch is activated. This sends a signal to the five second timer relay to operate the pump. The flowing water holds open the cartridge. When the water stops flowing the cartridge is held in the closed position to prevent water from spilling out during flight maneuvres. The cartridge self−vents when the system is drained.

Drain mast

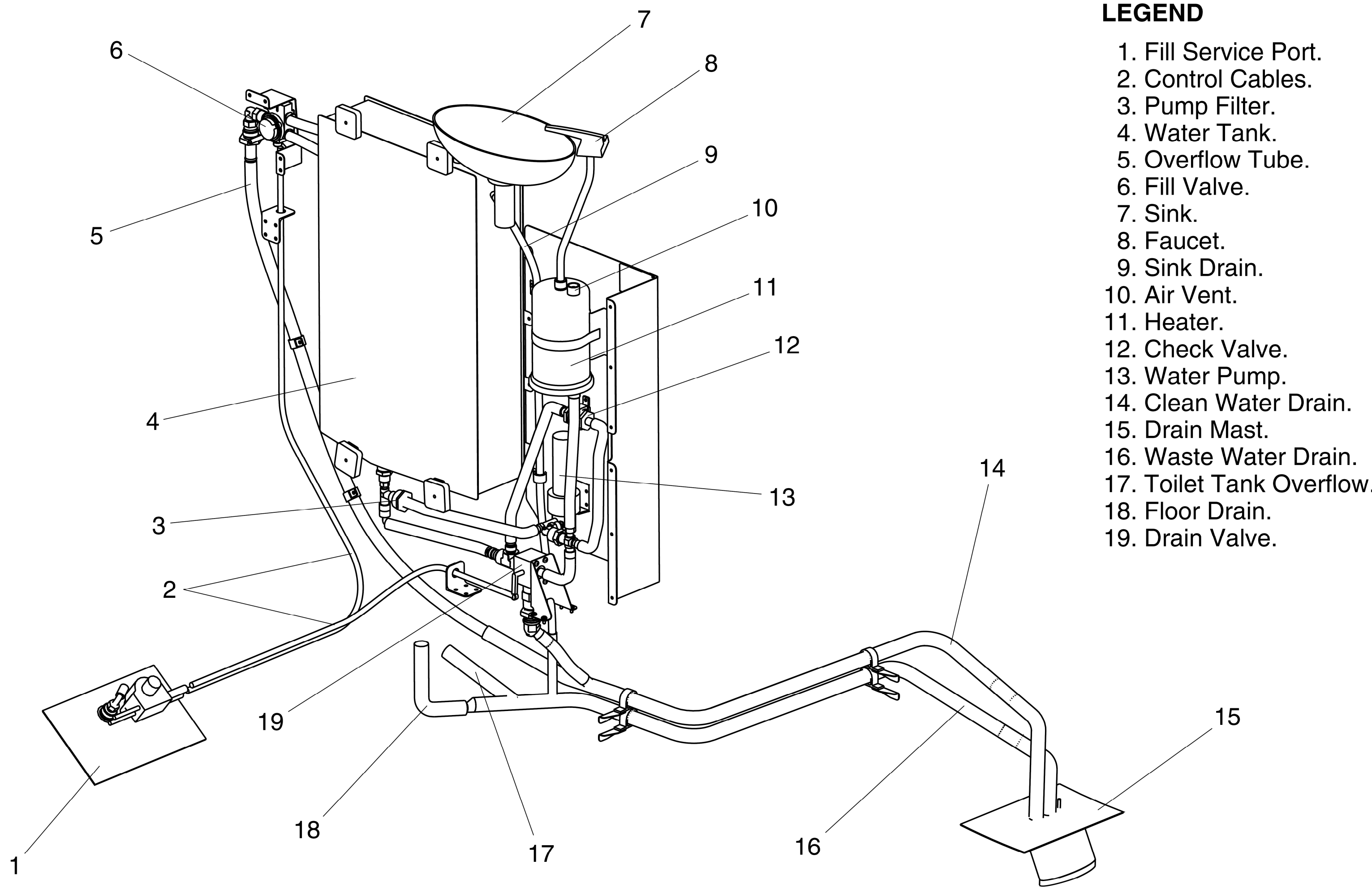
The drain mast has two drain lines attached to it. One line drains the waste water from the sink while the other line drains the clean water from the system. A fibreglass/silicon blanket with a built in heater covers the two lines.

A similar heater with a thermistor is attached to the drain mast. The two heating circuits are connected in parallel. The thermistor reduces the current to the heater as the temperature increases.

On aircraft with SB84−24−59 incorporated and without SB84−24−61, the drain mast circuit breaker located on the 115 V AC variable frequency circuit breaker panel is removed.

The ground service panel has a fill adapter, a T−handle and a cable control unit. The cable control unit has a control box installed behind the ground service panel with two cables coming out of it. One cable is connected to the fill valve and the other is connected to the drain valve.

Rotating the handle causes an inner cog in the control box to engage the cable and move it in or out. This action operates the fill valve connected to the cable and allows the system to be filled with water. Pulling the handle out disengages the inner cog and allows the handle to engage an outer cog. Rotating the handle then operates the drain valve in a similar way.



LEGEND

1. Fill Service Port.
2. Control Cables.
3. Pump Filter.
4. Water Tank.
5. Overflow Tube.
6. Fill Valve.
7. Sink.
8. Faucet.
9. Sink Drain.
10. Air Vent.
11. Heater.
12. Check Valve.
13. Water Pump.
14. Clean Water Drain.
15. Drain Mast.
16. Waste Water Drain.
17. Toilet Tank Overflow.
18. Floor Drain.
19. Drain Valve.

Figure 38-3 Lavatory Wash Water System – Detailed Components

Waste Water

Introduction

The lavatory waste water system supplies the water for the passenger washroom facilities on board the aircraft. The system also stores the waste water after passenger use and allows it to be removed from the aircraft through a service panel.

General Description

Refer Figure 38-4

The toilet unit is an electronically operated, recirculating flush toilet installed in the lavatory compartment. The lavatory compartment is manufactured in one piece. Any fluids spilled on to the floor are contained within the compartment and then drained overboard through the floor drain. The toilet unit is filled with clean water and is drained of waste water through an exterior service panel.

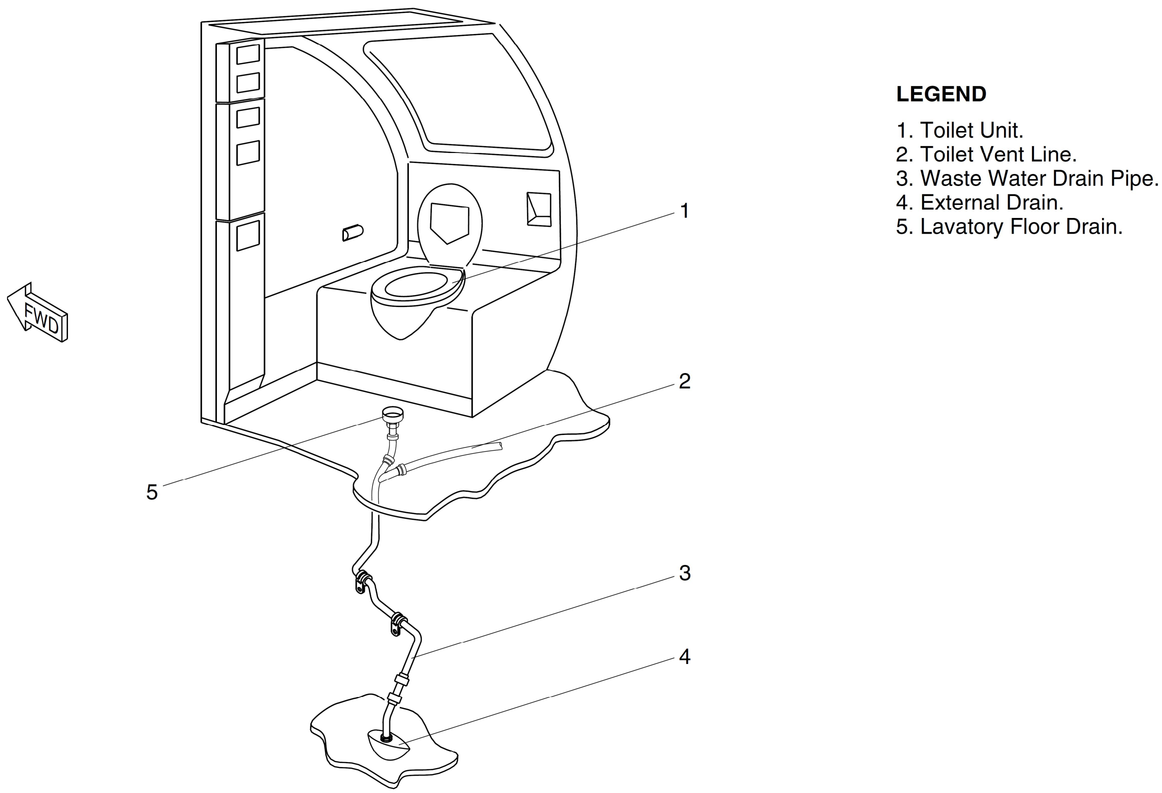


Figure 38-4 Lavatory System

Detailed Description

Refer Figure 38-5 & Refer Figure 38-6

The toilet unit has a reservoir tank which holds the fluid used for flushing as well as the waste fluid. The unit also has a motor pump cartridge with a built in filter basket, a timer unit, toilet bowl, flush/fill line with a check valve and a vent line with a muffler.

The toilet unit is attached to the lavatory compartment floor by hold down rods and is covered around the base by a decorative shroud.

The toilet bowl is made from polished stainless steel and is installed on top of the fluid/waste reservoir. A flush channel allows a flow of fluid to cleanse the bowl during flushing.

The unit is drained by dumping fluid and waste into a sewage drain which is directed to a service panel on the exterior of the aircraft.

Pushing the flush switch installed on the lavatory wall starts the flush cycle. When the flush switch is pushed, a timer unit installed on top of the toilet unit activates the 28 Vdc motor driven pump for 5 seconds.

The pump causes the flushing fluid to flow through the filter basket to the flush channel at the upper rim of the toilet bowl.

The flushing fluid flows down the inner side of the toilet bowl rinsing waste material directly into the reservoir.

A one–way hinged flapper valve prevents the waste material and fluid from flowing back into the bowl.

On aircraft with the Modsum 4–459505 incorporated, flush/fill cap is not installed.

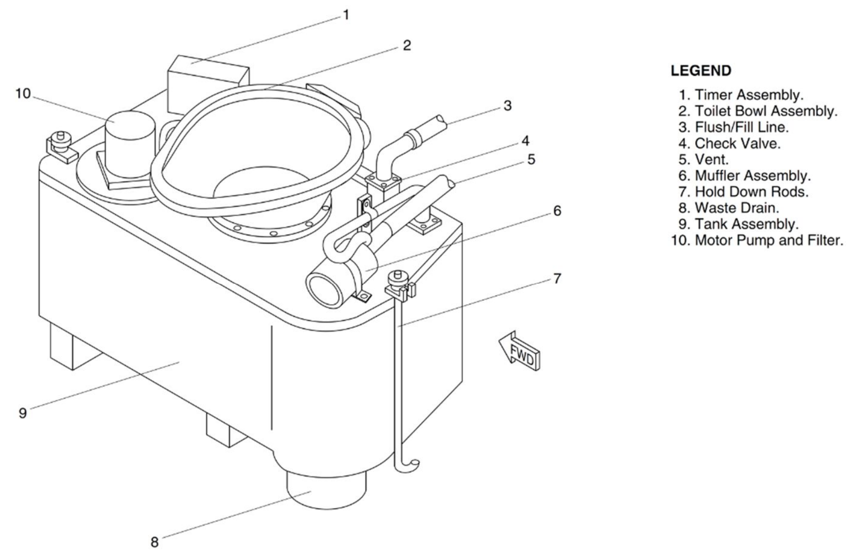


Figure 38-5 Lavatory Waste Disposal – Detailed

The lavatory waste water system is drained and filled through the lavatory service panel on the exterior of the aircraft. The sewage drain cover is opened and flush/fill cap removed.

The coupling connector and flush/fill hoses from the ground cart are connected to the appropriate connectors.

The sewage drain plug is removed by operating the plug extractor handle, which is part of the connector. The waste fluid and material drains into the ground service cart.

Once the tank has drained, the system is flushed with 10 gallons (37.85 liters) of pressurized water. The water enters the tank through the check valve to the pump inlet and a rotating spray nozzle.

The pressurized fluid sprays through the nozzle and cleans the inside of the filter basket and tank sidewalls. Then the drain plug is re–installed, and the unit is filled with 9.5 L of water and deodorant solution.

The drain connector and flush/fill line are then disconnected and the sewage drain cover and flush/fill cap are replaced.

The toilet flush electric motor is powered from the L MAIN bus through a 5 A circuit breaker. The electrical connector is attached to the timer unit.

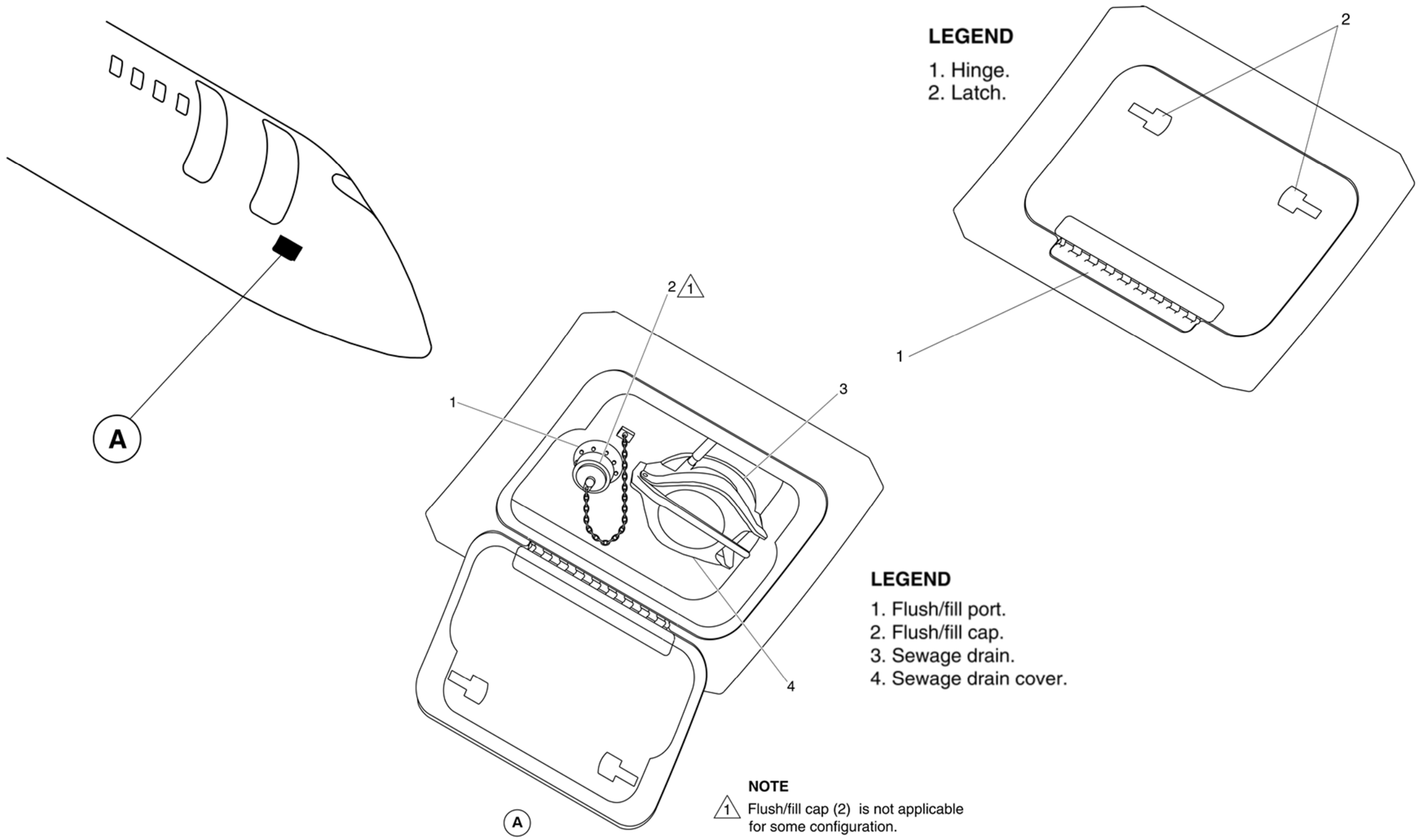


Figure 38-6 Lavatory Service Panel – Detailed

Waste Water - Component Description

Refer Figure 38-7

Bowl Assembly

The toilet bowl assembly is a polished stainless steel bowl installed on the top of the tank assembly. The shape of the bowl forces the water and waste flow into the tank.

A flapper valve at the bottom of the bowl prevents waste and water from re−entering the bowl from the tank.

Tank Assembly

The tank and top assembly are made from fiberglass, with a built in structural reinforcement. The top has provisions for the installation of the bowl assembly, the motor, pump and filter assembly, the timer, the flush/fill check valve and the vent line elbow.

Pump and Filter

The motor pump and filter is an electric pump/filter assembly installed in the top of the tank assembly. The filter is a TEF coated basket type filter that sits in the flush fluid.

The motor has a built in thermostat which disconnects power when the temperature reaches a preset level. This protects against overheating of the motor if it runs continuously.

Timer Assembly

The timer assembly is an electronic unit that is installed on top of the tank assembly. When the toilet flush timer switch is pushed, the timer unit will operate the pump for 5 seconds.

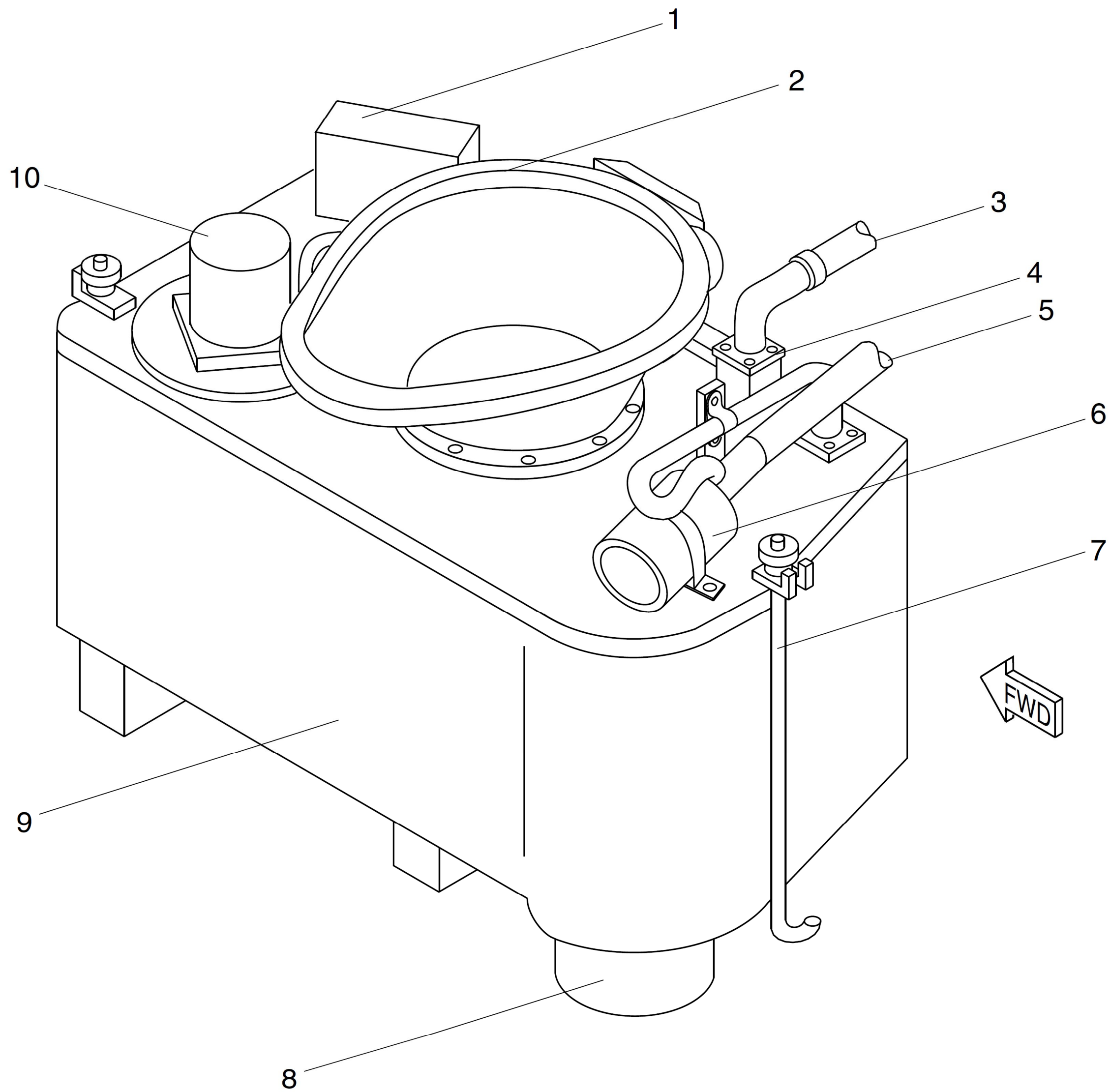
Check Valve

The flush/fill check valve is a mechanical valve installed in the top of the tank assembly. It allows water from the ground cart into the tank for cleaning and refilling, but does not allow water to flow back out through the fill tube.

Vent Line and Muffler

The tank is vented to the atmosphere through a vent line attached to the top of the tank assembly. This keeps toilet odors from entering the lavatory compartment. The vent line is open to the atmosphere and makes a constant noise while the aircraft is in operation.

A muffler assembly is installed in the vent line to reduce the noise.



LEGEND

1. Timer Assembly.
2. Toilet Bowl Assembly.
3. Flush/Fill Line.
4. Check Valve.
5. Vent.
6. Muffler Assembly.
7. Hold Down Rods.
8. Waste Drain.
9. Tank Assembly.
10. Motor Pump and Filter.

Figure 38-7 Lavatory Waste Disposal – Detailed Components